LAN JIN

Ph.D. Candidate | Plasmas & Nuclear Fusion | Ultrafast Photoemission & Computational Plasma Physics

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RESEARCH PROFILE

Ph.D. candidate in Nuclear Engineering and Radiological Sciences, majoring in Plasmas & Nuclear Fusion. Research integrates exact quantum and time-dependent Schrodinger-equation models, plasma simulation, and scientific machine learning to study electron emission, plasma startup, and fusion-relevant systems.

CURRENT RESEARCH

- **Ultrafast photoemission and electron dynamics:** Investigating quantum electron emission from metallic materials under ultrafast and strong optical fields, with emphasis on emission dynamics, energy-resolved spectra, quantum efficiency, pulse-shape effects, and carrier-envelope-phase control.
- **Fusion plasma modeling — Helion Energy HERCULES collaboration:** Modeling the formation and early evolution of preionized plasmas for fusion startup, including ionization dynamics, plasma formation, and edge behavior, as part of the University of Michigan–Helion HERCULES collaboration. Project overview
- Machine learning for computational physics: Developing surrogate and interpretable machine-learning models for computational photoemission and plasma systems to accelerate parameter-space exploration, identify nonlinear dependencies, and support physics-based model development. Co-authored atmospheric applications in turbulence-dissipation reconstruction and Doppler-lidar mixing-layer-height retrieval.

EDUCATION

Ph.D., Nuclear Engineering and Radiological Sciences **Jan 2025 - Present | Expected summer 2027**

University of Michigan, Ann Arbor, MI | Major/Research Area: Plasmas & Nuclear Fusion | Advisor: Prof. Peng Zhang

M.S., Electrical and Computer Engineering **2022 - 2024**

Michigan State University, East Lansing, MI | Advisor: Prof. Peng Zhang

M.S., Physical Electronics **2014 - 2017**

Yunnan University, Kunming, China

B.S., Fundamental Science of Mathematics and Physics **2010 - 2014**

Yunnan University, Kunming, China

RESEARCH EXPERIENCE

Summer Research — AI in the Nuclear Sector **Jun 22, 2026 - Aug 29, 2026**

University of Michigan | Nuclear Engineering and Radiological Sciences | Supervisor: Prof. Aditi Verma

- Investigated applications of artificial intelligence in nuclear safety, security, and safeguards, with emphasis on governance, regulation, and the historical development of AI within the nuclear sector.
- Contributed to the design and development of a survey for a nuclear-sector AI observatory, aimed at collecting data on the use, oversight, and emerging trends of AI in nuclear applications.

Graduate Research Assistant **Jan 2025 - Present**

University of Michigan | Nuclear Engineering and Radiological Sciences | Advisor: Prof. Peng Zhang

- Published first-author theory demonstrating orders-of-magnitude quantum-efficiency enhancement as laser pulses shorten to the few-cycle regime at fixed low intensity; quantified pulse-length, photon-order, and carrier-envelope-phase effects.
  - Advanced exact quantum and TDSE-based models for ultrafast photoemission to investigate electron-emission dynamics, energy-resolved spectra, emitted-electron properties, and quantum efficiency.
- Contributed to PASCHEN-1D, a one-dimensional fluid plasma model that incorporates photoemission, thermionic emission, secondary-electron emission, and RLC circuit coupling to study gas breakdown and Paschen-scaling behavior.
  - Developed ML surrogates using Scikit-learn, TensorFlow, and Keras for emission and plasma simulations; applied feature attribution to guide physics-based refinement.
- Co-authored two 2026 atmospheric-science papers on physics-guided machine learning for spatiotemporal turbulence-dissipation reconstruction and Doppler-lidar mixing-layer-height retrieval.

Graduate Research Assistant **Jan 2022 - Dec 2024**

Michigan State University | Electrical and Computer Engineering | Advisor: Prof. Peng Zhang

- An exact-quantum-model analysis of direct density modulation in photo-assisted field emission from an RF cold cathode, including current-pulse amplitude, width, shape, harmonic content, and modulation depth.
- Co-invented an optically gated field-emission cathode architecture; U.S. nonprovisional application filed Aug. 9, 2024 and currently patent pending.
- Presented research at APS-DPP, IEEE ICOPS, IVEC/IVESC, IVNC, GEC, and MIPSE; received the 2024 MIPSE Fellowship.

Co-trained Research Student **Aug 2015 - Jan 2017**

Institute of Physics, Chinese Academy of Sciences | Ultracold Atoms and Polar Molecules Group | Advisor: Prof. Ru-Quan Wang

- Contributed to a high-performance 2D+ magneto-optical trap that captured 40K from a natural-abundance source and supported degenerate 23Na/40K Bose-Fermi mixtures for ultracold polar-molecule production.
- Designed and implemented free-space electro-optical modulators for multiple drive frequencies and high-speed optical switching.

PEER-REVIEWED PUBLICATIONS

1. Chu, Y., Lin, G., Zhang, J. A., Xue, L., Li, W., Shin, H. H., Deng, M., Zhang, D., Guo, H., Wang, Z., and Jin, L. Surface-constrained reconstruction of spatiotemporal turbulence dissipation in the convective boundary layer using physics-guided machine learning. Journal of Geophysical Research: Atmospheres 131, e2026JD047285 (2026). DOI: 10.1029/2026JD047285
2. Chu, Y., Lin, G., Zhang, J. A., Xue, L., Li, W., Shin, H. H., Deng, M., Zhang, D., Guo, H., Li, Z., and Jin, L. A Physically-Guided and Stage-Wise Machine Learning Framework for Mixing Layer Height Retrieval Based on Doppler Lidar Observations. Advances in Atmospheric Sciences (2026). DOI: 10.1007/s00376-026-6100-y
3. Iqbal, A., Heri, Y., Wang, B., **Jin, L.**, Faisal, M. A., and Zhang, P. "PASCHEN-1D: A one-dimensional fluid plasma solver with multi-mechanism surface emission and flexible external circuit coupling." Computer Physics Communications 329, 110404 (2026). DOI: 10.1016/j.cpc.2026.110404

4. **Jin, L.**, and Zhang, P. "Enhanced quantum efficiency in photoemission driven by few-cycle optical fields." *Physical Review Research* 8, 013035 (2026). DOI: 10.1103/zhd1-5d6q
5. **Jin, L.**, Zhou, Y., and Zhang, P. "Direct density modulation of photo-assisted field emission from an RF cold cathode." *Journal of Applied Physics* 134, 074904 (2023). DOI: 10.1063/5.0156328
6. Yang, J.-L., Long, Y., Gao, W.-W., **Jin, L.**, Zuo, Z.-C., and Wang, R.-Q. "Enhanced loading of 40K from a natural-abundance potassium source with a high-performance 2D+ MOT." *Chinese Physics Letters* 35, 033701 (2018). DOI: 10.1088/0256-307X/35/3/033701

PATENTS

7. Zhang, P., **Jin, L.**, and Zhou, Y. "Optically Gated Field Emission Cathodes." U.S. patent application; provisional filed Aug. 14, 2023; nonprovisional filed Aug. 9, 2024; patent pending.
8. **Jin, L.**, and Zheng, J. "Display Panel and Display Equipment." Chinese Utility Model Patent CN201920865304.2; granted Jan. 7, 2020.
9. **Jin, L.**, and Yang, C. "Display Panel and Displayer." Chinese Utility Model Patent CN201920864409.2; granted Jan. 7, 2020.

SELECTED CONFERENCE & SYMPOSIUM PRESENTATIONS

2026 Jin, L., and Zhang, P. “Temporal shape effects of femtosecond laser pulses on photoemission from a metal surface.” 2026 IEEE ICOPS. [Oral]

2025 Jin, L., and Zhang, P. "Quantum efficiency of photoemission induced by few-cycle optical fields." 67th Annual Meeting of the APS Division of Plasma Physics. *[Oral]*

2025 Jin, L., and Zhang, P. "Photon-assisted thermionic emission: A theoretical prediction for photoelectron energy spectra, emission current, and QE." MIPSE Symposium, University of Michigan. *[Poster]*

2024 Slovak, A., Jin, L., Heri, Y., Mankowski, J., Dickens, J., Neuber, A., Zhang, P., and Stephens, J. "An apparatus for measuring laser-assisted field emission in a vacuum diode." IEEE IPMHVC, Indianapolis, IN. *[Poster]*

2024 Jin, L., and Zhang, P. "Effects of laser pulse length on photoemission spectra from a biased metal surface." Joint IVEC + IVESC. *[Oral]*

2024 Jin, L., and Zhang, P. "Influence of temporal shapes of femtosecond laser pulses on photoemission from a metal surface." 15th MIPSE Graduate Student Symposium. *[Poster]*

2023 Jin, L., Zhou, Y., and Zhang, P. "Density-modulated electron emission using RF and laser fields." IEEE ICOPS. *[Oral]*

2023 Jin, L., Zhou, Y., and Zhang, P. "Modulated electron beam emission under RF and laser fields." IEEE IVNC. *[Poster]*

2023 Jin, L., Zhou, Y., and Zhang, P. "Beam density modulation during emission under RF and laser fields." 76th Gaseous Electronics Conference. *[Poster]*

HONORS & AWARDS

2024 MIPSE Fellowship, Michigan Institute for Plasma Science and Engineering

2016 Yuehong Scholarship, Yunnan University

2015 Third Academic Scholarship, Yunnan University

2014 First Academic Scholarship, Yunnan University

2012 Outstanding Student Award (Third Prize), Yunnan University

INDUSTRY EXPERIENCE

TV Mask Design Engineer **Apr 2018 - Sep 2019**

Chuzhou HKC Photoelectric Technology Co., Ltd. | TFT-LCD volume manufacturing

- Designed photolithography masks for TFT-LCD display panels and collaborated with process, yield, and product engineering teams during volume manufacturing.
- Co-invented two display-panel innovations that received Chinese Utility Model Patents in 2020.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, C/C++, Mathematica

Quantum & Numerical Modeling: 1D TDSE, exact quantum transport/emission models, Poisson and drift-diffusion solvers, Fourier/spectral methods, finite-difference and finite-volume schemes

Plasma Simulation: Fluid and hybrid models, particle-in-cell methods, WarpX, CCP/ICP, EDIPIC, sheath physics, plasma-surface interaction, preionized-plasma and discharge modeling

Machine Learning: Scikit-learn, TensorFlow, Keras; regression, ensemble models, neural networks, surrogate modeling, feature attribution

Research Computing: Linux/HPC, Git, SLURM, LaTeX, reproducible scientific workflows

PROFESSIONAL MEMBERSHIPS

APS: Division of Plasma Physics, since 2025

IEEE: Member; Nuclear and Plasma Sciences Society, since 2022